



DC – AI – Routing – Switching – Security – Wireless – Automation

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An HPE Juniper Networking Techpost

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HPE



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Example settings for connecting a VPN from the native IKEv2 client on Android 13+ to a Juniper SRX firewall. Due to the client's nature, use cases may include basic remote access and embedded/IoT scenarios where additional software is undesirable.

0.1 Introduction

For connecting a VPN from the Android platform to the Juniper SRX, multiple options exist:

- Fully supported Juniper Secure Connect (JSC) [client available in Google Play](#)
- Add-on third-party IPsec app, with [strongSwan](#) Open Source being a typical choice
- Native [Android 11+ IKEv2 implementation](#) written in Java

This article focuses on the Android 11+ native client (Android 13+ practically with the SRX), where the most applicable scheme is IKEv2 mutual RSA certificate-based authentication, referred to in the client settings as IKEv2/IPSec RSA. Neither ECDSA nor EAP-TLS is supported in vanilla Android, as indicated by the scheme name. To support IKEv2 without EAP, the SRX side effectively uses a [pico-cell provisioning configuration](#), a VPN style originally designed for the initial enrolment of cell site equipment in mobile networks.

Note: There seem to be specific Android vendor implementations, e.g., Samsung, which supports EAP-TLS, as noted later.

The involved RADIUS backend is used by the SRX for authorization by looking up the username constructed from the client's IKE-ID with a statically defined password. In the case of mutual certificate authentication between the client and the SRX, the mapping of the certificate attribute to the IKE-ID (effectively the subject of authorization by the backend) is validated by the SRX based on [RFC 4945](#). In the Pre-Shared Keys scenario (referred to as IKEv2/IPSec PSK), a unique PSK per IKE-ID ensures the binding, which is ultimately authorized by RADIUS as well. The client IP addressing using the IKEv2 Configuration Payload (IKEv2 option to provision IP addresses and related settings from the responder to the initiator) for both authentication scenarios is either from a local address pool on the SRX or potentially provided as an attribute from the backend RADIUS, overriding the local pool when used.

Note: Naturally, a question related to VPNs from the Apple ecosystem arises. For the time being, there is an issue where iOS disconnects after approximately 24 minutes during the first re-key operation. The root cause is unknown and would likely require cooperation with a technical expert on Apple's IKEv2 implementation. The disconnect issue seems to be widespread based on internet searches. Please reach out if you happen to be a technical expert on Apple's IKEv2 implementation.

Although documented first, the IKEv2/IPSec PSK authentication is likely less suitable from a security perspective, particularly regarding user factors (such as PSK leaks) and manageability/scalability. Additionally, mixing the SRX PKI and PSK-based proposals for dynamic endpoints on the same interface is possible only when using an IP alias, as different IKE policies can't coexist for dynamic VPN endpoints. The provided configurations for the PSK scheme are based on the [SRX AutoVPN / PSK with Linux strongSwan](#) Tech Post article describing the SRX site-to-site AutoVPN for third-party spokes. For illustration purposes, the approach for both PSK and certificate schemes coexisting on the same device will be covered here.

In addition to certificate and PSK, the MSCHAPv2 scheme (which effectively involves EAP) is also implemented in the native Android client. However, it seems to have a known [padding issue](#) that is still present, at least in Android 13. The same SRX configuration with the strongSwan app in EAP-MSCHAPv2 mode works fine, leading to this conclusion.

For baseline purposes and troubleshooting (or even trying out without Android), the SRX IKEv2 PSK and certificate troubleshooting sections cover equivalent Linux strongSwan settings.

0.2 Android VPN Attributes

As of Android 13, the following attributes of the native Android IKEv2 implementation, in conjunction with the SRX, have been practically observed. The thoroughly tested devices include the Google Pixel 5 running Android 13-based [/e/OS](#) and the vanilla Pixel 7 running stock Android 14.

- IKE parameters suitable with the SRX (IKED IKE control plane):
 - AES-CBC 128/192/256, SHA-256/384/512
 - AES-GCM 128/256
 - CHACHA20-POLY1305
 - DH groups 14, 16
- Transform proposals (please note Android doesn't use Perfect Forward Secrecy):
 - AES-GCM 128/192/256
 - AES-CBC 128/192/256, SHA-256/384/512
 - CHACHA20-POLY1305
- IKE rekey timer of 1 hour, hardcoded
- IPSec rekey timer of 1 hour, hardcoded
- Dual stack support
 - Tunnel endpoint
 - IPv4 and IPv6 traffic selector, Configuration Payload client addressing
- Tunnel all policy only, no narrowing based on gateway traffic selectors
- IKEv2 fragmentation support
- Redial in case of connection change (e.g., mobile->WiFi)
- No tunnel MTU settings, reliance on gateway MSS and fragmentation handling
- Always-on option only with PSK/password scheme (Android 13); Android 14 also for PKI

Notes:

- Android 11 seems to be using HMAC-SHA1 as the Pseudo-Random Function (PRF) along with AES CBC/GCM ciphers, but the offered PRF is not compatible with the SRX.
- Android 12 (on Samsung S9 with [/e/OS](#)) has issues likely related to certificate requests within IKE, resulting in ID verification failures; this issue is not yet fully understood.
- Android 14 on the Samsung Galaxy S21 and Galaxy Tablet S6 Lite supports EAP-TLS; however, for example, DH group 16 is not available, but DH groups 19, 20, and 21 are. Additionally, IPv6 Configuration Payload addressing is not occurring, which seems to be specific to Samsung.

0.3 Demo Topology

Let's consider the following IPv4/IPv6 dual-stack topology with three zones: UNTRUST, TRUST and VPN zone accommodating tunnel interfaces for both certificate and PSK authentication. As mentioned in the introduction, dynamic endpoint IKE gateways with different proposals (certificate vs. PSK authentication) can't coexist on the same IP address; therefore, an additional pair of IP addresses exists for both public IPv4 and global IPv6. Config Payload IP pools are common (and the clients are effectively dual-stacked).

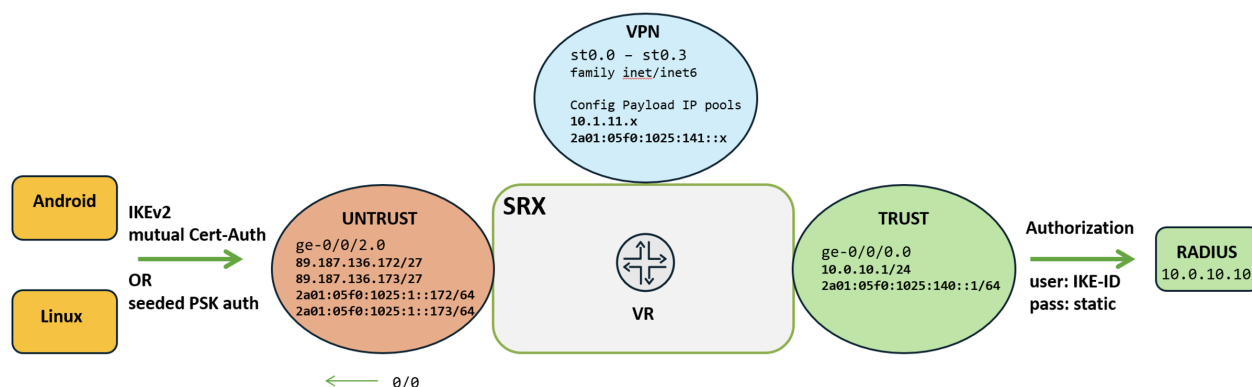


Figure 1: Topology

For better clarity, the following table outlines the mapping of external IP addresses to the tunnel adapters and authentication schemes.

External IP address	Related Tunnel interface	Common AutoVPN suffix
89.187.136.172	st0.0	@psk.auth
2a01:05f0:1025:1::172	st0.1	@psk.auth
89.187.136.173	st0.2	@cert.auth
2a01:05f0:1025:1::173	st0.3	@cert.auth

0.4 Sample RADIUS Settings

As noted above, the prerequisite for using the pico-cell VPN style is RADIUS for authorization purposes. The following is a sample of the RADIUS client definition on FreeRADIUS, running on a Debian 12 host (apt install freeradius) with an IP address of **10.0.10.10** from the topology drawing above (SRX as the RADIUS client **10.0.10.1**):

```
1 /etc/freeradius/3.0/clients.conf:
2
3 client vsrx {
4     ipaddr = 10.0.10.1
5     secret = juniper123
6 }
```

Then, the following is a simple per-user authorization of the IKE-ID, where the password is defined in the SRX configuration as config-payload-password setting (entire configuration in the SRX configuration sections):

```

1 /etc/freeradius/3.0/users:
2
3 user1@psk.auth Cleartext-Password := "juniper123"
4 user1@cert.auth Cleartext-Password := "juniper123"

```

The above method may not be ideal for scalability; however, options such as backend authorization using LDAP or SQL are available. For simple use cases, the following bulk authorization works for IKE-IDs: *@cert.auth* and *@psk.auth*.

```

1 /etc/freeradius/3.0/sites-enabled/default:
2
3 authorize {
4
5     if (User-Name =~ /. *@cert\.auth$/ && User-Password == "juniper123") {
6         update control {
7             Auth-Type := Accept
8         }
9     }
10
11     if (User-Name =~ /. *@psk\.auth$/ && User-Password == "juniper123") {
12         update control {
13             Auth-Type := Accept
14         }
15     }
16
17     <OTHER FreeRADIUS AUTHORIZE CONFIGURATION STANZAS>
18
19 }

```

After changing the configuration, a restart or reload of FreeRADIUS is needed:

```

1 systemctl reload freeradius

```

For troubleshooting, the following command provides verbose debug output to the terminal:

```

1 systemctl stop freeradius
2 freeradius -X

```

In the case of certificate authentication, clients can be effectively banned from connecting if their user certificate is revoked. Alternatively, for both certificate and PSK authentication, blocking certain IKE IDs is possible, as described in the Blocking Certain IKE IDs on the SRX chapter of the [SRX AutoVPN / PSK with Linux strongSwan](#) Tech Post.

Note: If the RADIUS server is EAP-TLS enabled, the SRX behaves transparently and can effectively serve as a VPN termination point for both mutual certificate authentication (with RADIUS authorization) and EAP-TLS authentication (e.g., strongSwan based clients).

0.5 Common SRX VPN Settings

As a prerequisite, IKED must be deployed. By default, for this Tech Post, the vSRX 3.0 used includes the older KMD IKE control daemon. The corresponding command for vSRX and the SRX 4000 series is:

```

1 > request system software add optional://junos-ike.tgz

```

Sidenote: Platforms such as the SRX1600, SRX2300, SRX4300, and SPC3 already have IKED installed, while branch SRX models do not support IKED. To check if IKED is installed:

```

1 > show version | match ike
2 JUNOS ike [20241219.060016_builder_junos_244_r1]

```

The idea behind the configuration approach is to separate the common elements used by both PSK and certificate authentication into distinct Junos groups. The complete configuration is provided in Appendix 1.

Below is the common group that expands on the basic settings in the top configuration hierarchy. Common settings include tuning of MSS, expansion of existing NAT rules, security policies:

```

1 edit groups dial-up-vpn-common
2
3 set security flow tcp-mss ipsec-vpn mss 1400
4 set security nat source rule-set untrust from zone vpn
5
6 set security policies from-zone vpn to-zone trust policy permit-vpn match source-address any
7 set security policies from-zone vpn to-zone trust policy permit-vpn match destination-address
  any
8 set security policies from-zone vpn to-zone trust policy permit-vpn match application junos-icmp
  -ping
9 set security policies from-zone vpn to-zone trust policy permit-vpn then permit
10 set security policies from-zone vpn to-zone untrust policy permit-vpn match source-address any
11 set security policies from-zone vpn to-zone untrust policy permit-vpn match destination-address
  any
12 set security policies from-zone vpn to-zone untrust policy permit-vpn match application any
13 set security policies from-zone vpn to-zone untrust policy permit-vpn then permit
14
15 set security zones security-zone vpn tcp-rst
16 set security zones security-zone untrust interfaces ge-0/0/2.0 host-inbound-traffic system-
  services ike

```

RADIUS, and client IP addressing (IKEv2 Config Payload for both IPv4 and IPv6):

```

1 set access profile ikev2-access-profile-1 authentication-order radius
2 set access profile ikev2-access-profile-1 address-assignment pool ikev2-address-pool-1
3 set access profile ikev2-access-profile-1 address-assignment inet6-pool ikev2-address-pool-2
4 set access profile ikev2-access-profile-1 radius-server 10.0.10.10 secret juniper123
5 set access profile ikev2-access-profile-1 radius-server 10.0.10.10 routing-instance vr
6
7 set access address-assignment pool ikev2-address-pool-1 family inet network 10.1.11.0/24
8 set access address-assignment pool ikev2-address-pool-1 family inet range ikev2-range-1 low
  10.1.11.2
9 set access address-assignment pool ikev2-address-pool-1 family inet range ikev2-range-1 high
  10.1.11.200
10 set access address-assignment pool ikev2-address-pool-1 family inet xauth-attributes primary-dns
  1.1.1.1/32
11 set access address-assignment pool ikev2-address-pool-1 family inet xauth-attributes secondary-
  dns 1.0.0.1/32
12 set access address-assignment pool ikev2-address-pool-2 family inet6 prefix 2a01:05f0
  :1025:141::/64
13 set access address-assignment pool ikev2-address-pool-2 family inet6 range ikev2-range-2 low 2
  a01:05f0:1025:141::2/128
14 set access address-assignment pool ikev2-address-pool-2 family inet6 range ikev2-range-2 high 2
  a01:05f0:1025:141::ff/128
15 set access address-assignment pool ikev2-address-pool-2 family inet6 xauth-attributes primary-
  dns-ipv6 2606:4700:4700::1111
16 set access address-assignment pool ikev2-address-pool-2 family inet6 xauth-attributes secondary-
  dns-ipv6 2606:4700:4700::1001

```

0.6 SRX IKEv2 PSK auth

0.6.1 SRX IKEv2 PSK Settings

In the PSK scenario, the only reasonable option is to use seeded PSKs, as this ensures that each user (with a unique IKE-ID) has a distinct PSK calculated based on an irreversible function, where the input consists of a PSK seed, IKE-ID, and salt.

The definition of proposals includes timers configured so that the client always initiates rekeying. Note the seeded-pre-shared-key stanza:

```

1 edit groups dial-up-vpn-psk
2
3 set security ike proposal ikev2-proposal-psk authentication-method pre-shared-keys
4 set security ike proposal ikev2-proposal-psk dh-group group16
5 set security ike proposal ikev2-proposal-psk encryption-algorithm aes-256-gcm
6 set security ike proposal ikev2-proposal-psk lifetime-seconds 57600
7 set security ike policy ikev2-policy-psk proposals ikev2-proposal-psk

```

```
8 set security ike policy ikev2-policy-psk seeded-pre-shared-key ascii-text juniper123
```

Due to how IKE gateways are handled by the SRX, there needs to be a distinct gateway for IPv4 and IPv6 endpoints. Below is the IPv4 configuration, which binds the public IP **89.187.136.172** and accepts IKE-IDs that include @psk.auth as the common part.

```
1 set security ike gateway ikev2-gw-ipv4-psk ike-policy ikev2-policy-psk
2 set security ike gateway ikev2-gw-ipv4-psk dynamic user-at-hostname "@psk.auth"
3 set security ike gateway ikev2-gw-ipv4-psk dynamic ike-user-type group-ike-id
4 set security ike gateway ikev2-gw-ipv4-psk local-identity hostname p4.jnpr.cz
5 set security ike gateway ikev2-gw-ipv4-psk external-interface ge-0/0/2.0
6 set security ike gateway ikev2-gw-ipv4-psk local-address 89.187.136.172
7 set security ike gateway ikev2-gw-ipv4-psk aaa access-profile ikev2-access-profile-1 config-
  payload-password juniper123
8 set security ike gateway ikev2-gw-ipv4-psk version v2-only
9 set security ike gateway ikev2-gw-ipv4-psk fragmentation size 1280
```

And correspondingly, the IPv6 gateway is configured with the address **2a01:5f0:1025:1::172**:

```
1 set security ike gateway ikev2-gw-ipv6-psk ike-policy ikev2-policy-psk
2 set security ike gateway ikev2-gw-ipv6-psk dynamic user-at-hostname "@psk.auth"
3 set security ike gateway ikev2-gw-ipv6-psk dynamic ike-user-type group-ike-id
4 set security ike gateway ikev2-gw-ipv6-psk local-identity hostname p4.jnpr.cz
5 set security ike gateway ikev2-gw-ipv6-psk external-interface ge-0/0/2.0
6 set security ike gateway ikev2-gw-ipv6-psk local-address 2a01:5f0:1025:1::172
7 set security ike gateway ikev2-gw-ipv6-psk aaa access-profile ikev2-access-profile-1 config-
  payload-password juniper123
8 set security ike gateway ikev2-gw-ipv6-psk version v2-only
9 set security ike gateway ikev2-gw-ipv6-psk fragmentation size 1280
```

IPsec (transform) proposal, please note that, based on observations, Android does not support Perfect Forward Secrecy (PFS):

```
1 set security ipsec proposal ipsec-proposal-psk protocol esp
2 set security ipsec proposal ipsec-proposal-psk encryption-algorithm aes-256-gcm
3 set security ipsec proposal ipsec-proposal-psk lifetime-seconds 3900
4 set security ipsec policy ipsec-policy-psk proposals ipsec-proposal-psk
```

VPN configuration for the IPv4 endpoint, based on observations, Android does not accept any remote traffic selectors other than **0.0.0.0/0** and **::/0**, respectively:

```
1 set security ipsec vpn ipsec-vpn-ipv4-psk bind-interface st0.0
2 set security ipsec vpn ipsec-vpn-ipv4-psk ike gateway ikev2-gw-ipv4-psk
3 set security ipsec vpn ipsec-vpn-ipv4-psk ike ipsec-policy ipsec-policy-psk
4 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
5 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
6 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv6 local-ip ::/0
7 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv6 remote-ip ::/0
```

IPv6 configuration corresponding to the above IPv4 VPN:

```
1 set security ipsec vpn ipsec-vpn-ipv6-psk bind-interface st0.1
2 set security ipsec vpn ipsec-vpn-ipv6-psk ike gateway ikev2-gw-ipv6-psk
3 set security ipsec vpn ipsec-vpn-ipv6-psk ike ipsec-policy ipsec-policy-psk
4 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
5 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
6 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv6 local-ip ::/0
7 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv6 remote-ip ::/0
```

The assignment of tunnel interfaces for both IPv4 and IPv6 to zones and routing instances is as follows:

```
1 set security zones security-zone vpn interfaces st0.0
2 set security zones security-zone vpn interfaces st0.1
3
4 set interfaces st0 unit 0 description ipsec-vpn-ipv4-psk
5 set interfaces st0 unit 0 family inet
6 set interfaces st0 unit 0 family inet6
7 set interfaces st0 unit 1 description ipsec-vpn-ipv6-psk
8 set interfaces st0 unit 1 family inet
9 set interfaces st0 unit 1 family inet6
10
```

```
11 set routing-instances vr interface st0.0
12 set routing-instances vr interface st0.1
```

Finally, apply both common and PSK groups:

```
1 top
2 set apply-groups dial-up-vpn-common
3 set apply-groups dial-up-vpn-psk
4 commit
```

0.6.2 Android IKEv2 PSK Validation

The first step in validating this scenario is to retrieve the per-user unique PSK. As long as the IKE policy remains the same, the key will be identical for both the IPv6 and IPv4 gateways:

```
1 > show security ike pre-shared-key gateway-name ikev2-gw-ipv4-psk user-id user1@psk.auth
2   Pre-shared key: b69d81f4c3e608a83d882c7aed8266fd88144b63
```

Client-side settings are straightforward; the Server address must match the one configured as local on the SRX and IPSec identifier is the one used for generating PSK:

← +

Edit VPN profile

Name

P4-PSK

Type

IKEv2/IPSec PSK ▼

Server address

p4.jnpr.cz

IPSec identifier

user1@psk.auth

IPSec pre-shared key

.....

☐ Show advanced options

☐ Always-on VPN

Forget Cancel Save

Status upon connection:

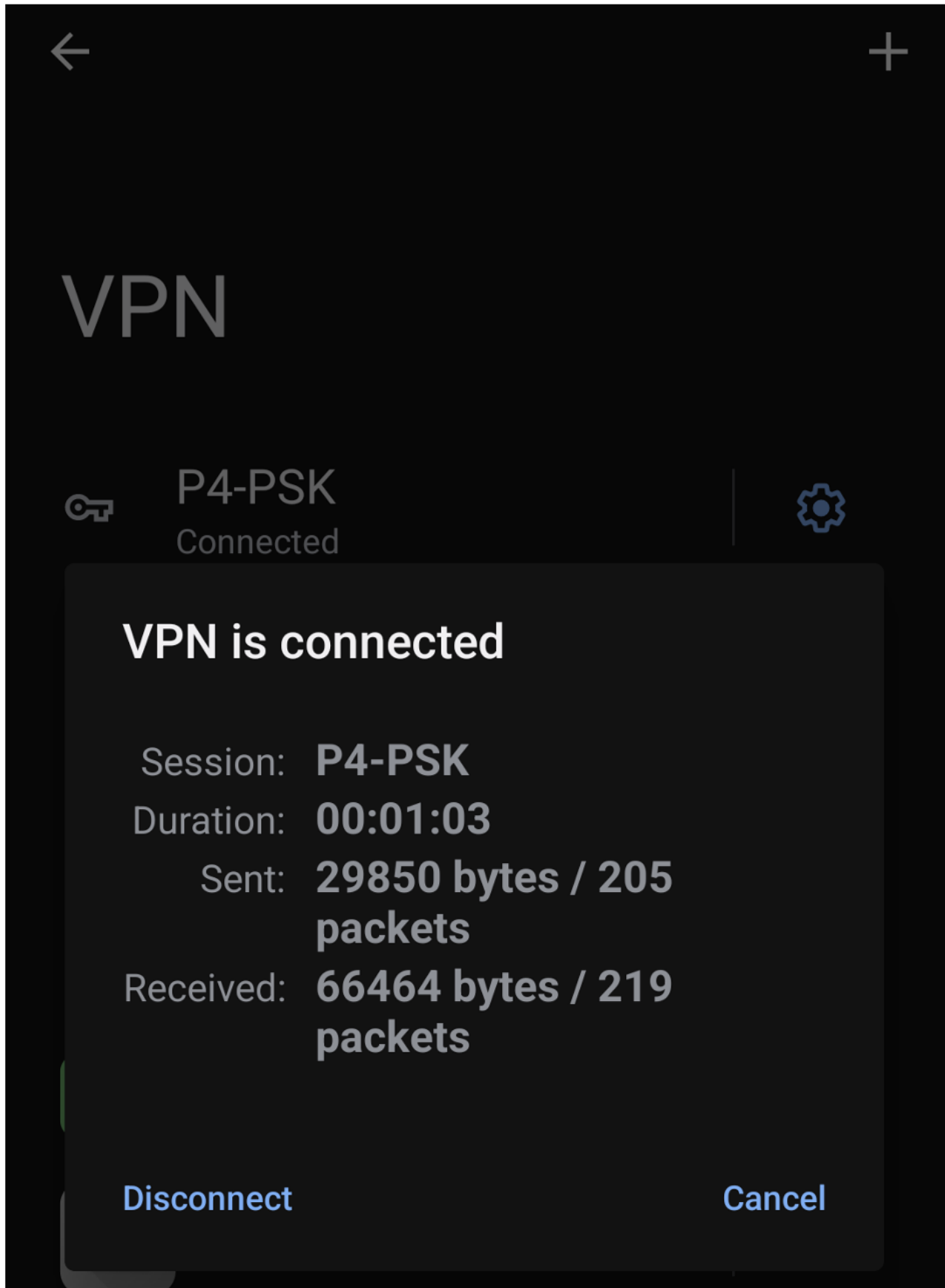


Figure 3: Status of Connected VPN

SRX side upon successful connection:

```

1 > show security ike active-peer detail
2
3
4 Local gateway interface: ge-0/0/2.0
5 Routing instance: vr
6 Local address: 89.187.136.172, Port: 4500,
7 Local IKE-ID : p4.jnpr.cz
8 Peer address: 81.93.249.121, Port: 5928,
9 Peer IKE-ID : user1@psk.auth
10 AAA username: not available
11 Assigned network attributes:
12 IP Address : 10.1.11.18 , netmask : 255.255.255.0
13 DNS Address : 1.1.1.1 , DNS2 Address : 1.0.0.1
14 WINS Address : 0.0.0.0 , WINS2 Address : 0.0.0.0
15 Assigned network attributes (IPv6):
16 IP Address : 2a01:5f0:1025:141::12 , prefix : 64
17 DNS Address : 2606:4700:4700::1111 , DNS2 Address : 2606:4700:4700::1001
18
19 Previous Peer address : 0.0.0.0, Port : 0
20 Active IKE SA indexes : 5974
21 IKE SA negotiated : 1
22 IPSec tunnels active : 1, IPSec Tunnel IDs : 500027

```

Corresponding IKED logs are as follows:

```

1 vsrx-p4 iked[17437]: IKE_GATEWAY_ID_VALIDATION_SUCCESS: IKE: Gateway N:ikev2-gw-ipv4-psk L
:89.187.136.172:500 R:81.93.249.121:5928 Successful ike-id:user1@psk.auth U:N/A IKE:IKEv2
Role:R
2 \vrsrx-p4 iked[17437]: IKE_VPN_UP_ALARM_USER: VPN ipsec-vpn-ipv4-psk from 81.93.249.121 is up.
Local-ip: 89.187.136.172, gateway name: ikev2-gw-ipv4-psk, vpn name: ipsec-vpn-ipv4-psk,
tunnel-id: 500027, local tunnel-if: st0.0, remote tunnel-ip: 80.92.248.120, Local IKE-ID: p4
.jnpr.cz, Remote IKE-ID: user1@psk.auth, AAA username: Not-applicable, VR id: root, Traffic-
selector: ts1, Traffic-selector local ID: ipv4(0,0-65535,0.0.0.0-255.255.255.255) ipv6
(0,0-65535,::-ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff) , Traffic-selector remote ID: ipv4
(0,0-65535,10.1.11.18-10.1.11.18) ipv6(0,0-65535,2a01:5f0:1025:141::12-2a01:5f0
:1025:141::12) , SA Type: dynamic

```

0.6.3 SRX IKEv2 PSK Troubleshooting

An easy way to verify if the SRX configuration is correct is to test it from a Linux client, which provides rich debugging capabilities. One of the available options is the [strongSwan](#) OSS software, which is included in modern Linux distributions.

To install the minimal strongSwan packages on Debian Linux 12, the following list of packages can be used for easy copy/pasting as an argument of the apt install command (this will bring in the necessary dependencies):

```

1 charon-systemd libstrongswan libstrongswan-extra-plugins libstrongswan-standard-plugins
strongswan-libcharon strongswan-swanctl

```

The next step is to generate the Linux-specific (seeded) PSK on the SRX for user2@psk.auth:

```

1 > show security ike pre-shared-key gateway-name ikev2-gw-ipv4-psk user-id user2@psk.auth
2 Pre-shared key: 724ce42300e0d3f65ab739dababde2eeb7d8559a

```

Next, the example settings for the IKEv2 VPN in the Linux /etc/swanctl/swanctl.conf file for the specific setup with IPv4 and IPv6 Config Payload addressing are as follows (the remote IPv4 selector is set to specific prefixes to avoid potential loss of management connection when using **0.0.0.0/0**).

```

1 connections {
2     p4 {
3         remote_addrs = p4.jnpr.cz
4         version = 2
5         proposals = aes256gcm16-prfsha384-modp4096
6         keyingtries = 0

```

```

7     dpd_delay = 30
8     rekey_time = 8h
9     vips= 0.0.0.0, ::
10    local {
11        auth = psk
12        id = user2@psk.auth
13    }
14    remote {
15        auth = psk
16        id = p4.jnpr.cz
17    }
18    children {
19        p4 {
20            remote_ts = 10.0.10.0/24, 1.1.1.1/32, ::/0
21            esp_proposals = aes256gcm16
22            start_action = none
23            close_action = none
24            dpd_action = none
25        }
26    }
27 }
28 }
29 secrets {
30     ike-srx {
31         id = p4.jnpr.cz
32         secret = 724ce42300e0d3f65ab739dababde2eeb7d8559a
33     }
34 }

```

Finally, restart strongSwan and initiate the VPN. The terminal output will indicate a successful VPN connection, followed by a listing of the IPsec Security Associations (SAs):

```

1 # systemctl restart strongswan
2 # swanctl --initiate --child p4
3
4 [IKE] initiating IKE_SA p4[1] to 89.187.136.172
5 [ENC] generating IKE_SA_INIT request 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(
6   HASH_ALG) N(REDIR_SUP) ]
7 [NET] sending packet: from 192.168.123.121[500] to 89.187.136.172[500] (712 bytes)
8 [NET] received packet: from 89.187.136.172[500] to 192.168.123.121[500] (786 bytes)
9 [ENC] parsed IKE_SA_INIT response 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N((40002))
10   V V V V ]
11 [IKE] received NetScreen Technologies vendor ID
12 [ENC] received unknown vendor ID: 09:00:26:89:df:d6:b7:12
13 [IKE] received Juniper SRX vendor ID
14 [ENC] received unknown vendor ID: 67:96:70:50:f1:38:eb:51:de:0c:a6:5f:54:f1:f0:36
15 [CFG] selected proposal: IKE:AES_GCM_16_256/PRF_HMAC_SHA2_384/MODP_4096
16 [IKE] local host is behind NAT, sending keep alive
17 [IKE] sending cert request for "CN=P1_CA"
18 [IKE] authentication of 'user2@psk.auth' (myself) with pre-shared key
19 [IKE] establishing CHILD_SA p4{1}
20 [ENC] generating IKE_AUTH request 1 [ IDi N(INIT_CONTACT) CERTREQ IDr AUTH CPRQ(ADDR ADDR6 DNS
21   DNS6) SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(EAP_ONLY) N(MSG_ID_SYN_SUP) ]
22 [NET] sending packet: from 192.168.123.121[4500] to 89.187.136.172[4500] (422 bytes)
23 [NET] received packet: from 89.187.136.172[4500] to 192.168.123.121[4500] (428 bytes)
24 [ENC] parsed IKE_AUTH response 1 [ IDr AUTH CPRP(ADDR MASK DNS ADDR6 DNS6 DNS6) SA TSi TSr ]
25 [IKE] authentication of 'p4.jnpr.cz' with pre-shared key successful
26 [CFG] handling INTERNAL_IP4_NETMASK attribute failed
27 [IKE] installing DNS server 1.1.1.1 to /etc/resolv.conf
28 [IKE] installing DNS server 1.0.0.1 to /etc/resolv.conf
29 [IKE] installing DNS server 2606:4700:4700::1111 to /etc/resolv.conf
30 [IKE] installing DNS server 2606:4700:4700::1001 to /etc/resolv.conf
31 [IKE] installing new virtual IP 10.1.11.4
32 [IKE] installing new virtual IP 2a01:5f0:1025:141::4
33 [IKE] IKE_SA p4[1] established between 192.168.123.121[user2@psk.auth]...89.187.136.172[p4.jnpr.
34   cz]
35 [IKE] scheduling rekeying in 28336s
36 [IKE] maximum IKE_SA lifetime 31216s
37 [CFG] selected proposal: ESP:AES_GCM_16_256/NO_EXT_SEQ
38 [IKE] CHILD_SA p4{1} established with SPIs c0d2b836_i 720d220a_o and TS 10.1.11.4/32 2a01:5f0
39   :1025:141::4/128 == 1.1.1.1/32 10.0.10.0/24 ::/0
40 initiate completed successfully
41
42 # swanctl --list-sas
43
44 p4: #1, ESTABLISHED, IKEv2, 169735011a7aab7c_i* da9daeb76f214ae9_r
45   local 'user2@psk.auth' @ 192.168.123.121[4500] [10.1.11.4 2a01:5f0:1025:141::4]
46   remote 'p4.jnpr.cz' @ 89.187.136.172[4500]
47   AES_GCM_16-256/PRF_HMAC_SHA2_384/MODP_4096

```

```

43 established 2s ago, rekeying in 28334s
44 p4: #1, reqid 1, INSTALLED, TUNNEL-in-UDP, ESP:AES_GCM_16-256
45 installed 2s ago, rekeying in 3243s, expires in 3958s
46 in c0d2b836, 0 bytes, 0 packets
47 out 720d220a, 0 bytes, 0 packets
48 local 10.1.11.4/32 2a01:5f0:1025:141::4/128
49 remote 1.1.1.1/32 10.0.10.0/24 ::/0

```

0.7 SRX IKEv2 Certificate auth

The obvious prerequisite for certificate authentication is a PKI infrastructure that issues certificates to SRX VPN head-end and remote access clients. In an EAP-TLS scenario, a certificate for RADIUS must also be issued. However, in our specific example of mutual certificate authentication, RADIUS serves solely as authorization for IKE-IDs, without the use of EAP.

0.7.1 PKI Infrastructure

There are professional PKI solutions and various wrappers, primarily around OpenSSL. For this specific example, a tailored wrapper script named [SRX_VPN_demo_CA](#) will be used for SRX VPN demonstration purposes (reading the online documentation is recommended). The following are the steps conducted on Debian Linux 12 to set up a Certificate Authority (CA), the SRX VPN head-end certificate, and user certificates (for Android VPN and Linux troubleshooting purposes).

To install the SRX VPN demo CA from GitHub, either cloning the repository or download of the .zip file directly from GitHub is possible. To clone:

```

1 apt install --no-install-recommends git
2 git clone https://github.com/JNPRAutomate/SRX_VPN_demo_CA
3 cd SRX_VPN_demo_CA
4 chmod 755 .needed_binaries.sh *.sh

```

Generation of CA key/certificate valid for 10 years (by default, 4096-bit RSA with SHA-256) with a common name of VPN_CA:

```

1 ./1_CA_init.sh VPN_CA 3650

```

FQDN (Fully Qualified Domain Name) for the certificate's DNS Subject Alternative Name of the SRX head-end, followed by the generation of a key/certificate valid for 2 years:

```

1 echo p5.jnpr.cz > cert_list_server
2 ./2_CA_gencerts_from_cert_list-server.sh 730

```

Configuration of the sample password (pass) for PKCS#12 containers containing the user key and certificate, along with the specification of the user's FQDN (email) Subject Alternative Names (SAN) for matching the AutoVPN common IKE-ID part. This is followed by the generation of certificates valid for 1 year:

```

1 echo pass > PKCS12_password
2 echo user1@cert.auth > cert_list_user
3 echo user2@cert.auth >> cert_list_user
4 ./3_CA_gencerts_from_cert_list-user.sh 365

```

Sample transfer of the private key, certificate, and CA certificate to the SRX, along with the transfer of PKCS#12 containers and the CA certificate from the CA host machine to the distribution site (ideally the private keys would be generated on-box):

```

1 scp CA/cacert.pem CA/p5_jnpr_cz-key.pem CA/p5_jnpr_cz-cert.pem vsrx:
2 scp CA_host:SRX_VPN_demo_CA/CA/{user*.p12, cacert.crt} .

```

0.7.2 SRX Side PKI Setup

The CA profile is intentionally kept outside of the Junos group because if the group is deactivated, the CA certificate would be cleared. Therefore, the CA profile is configured in the top config hierarchy:

```
1 set security pki ca-profile VPN_CA ca-identity VPN_CA
2 set security pki ca-profile VPN_CA revocation-check disable
```

The next step is to load the CA certificate and private key along with matching certificate, followed by validation. Typical failure reasons are related to the CRL (or generally revocation check), which is deactivated for demonstration purposes:

```
1 > request security pki local-certificate load filename p5_jnpr_cz-cert.pem key p5_jnpr_cz-key.
   pem certificate-id p5-jnpr_cz_VPN_CA
2 > request security pki ca-certificate load ca-profile VPN_CA filename cacert.pem
3 > request security pki local-certificate verify certificate-id p5-jnpr_cz_VPN_CA
```

0.7.3 SRX IKEv2 Cert Settings

Definition of proposals with timers configured so that the client always initiates rekeying. The difference between the PSK proposal and the certificate-based proposal lies in the use of different authentication methods (RSA signatures) and the reference to the certificate loaded in the previous step:

```
1 edit groups dial-up-vpn-cert
2
3 set security ike proposal ikev2-proposal-cert authentication-method rsa-signatures
4 set security ike proposal ikev2-proposal-cert dh-group group16
5
6 set security ike proposal ikev2-proposal-cert encryption-algorithm aes-256-gcm
7 set security ike proposal ikev2-proposal-cert lifetime-seconds 57600
8
9 set security ike policy ikev2-policy-cert proposals ikev2-proposal-cert
10 set security ike policy ikev2-policy-cert certificate local-certificate p5-jnpr_cz_VPN_CA
```

Due to how IKE gateways are handled by the SRX, there needs to be a distinct gateway for IPv4 and IPv6 endpoints. Below is the IPv4 configuration, which binds the public IP 89.187.136.173 and accepts the common IKE-ID part @cert.auth:

```
1 set security ike gateway ikev2-gw-ipv4-cert ike-policy ikev2-policy-cert
2 set security ike gateway ikev2-gw-ipv4-cert dynamic user-at-hostname "@cert.auth"
3 set security ike gateway ikev2-gw-ipv4-cert dynamic ike-user-type group-ike-id
4 set security ike gateway ikev2-gw-ipv4-cert local-identity hostname p5.jnpr.cz
5 set security ike gateway ikev2-gw-ipv4-cert external-interface ge-0/0/2.0
6 set security ike gateway ikev2-gw-ipv4-cert local-address 89.187.136.173
7 set security ike gateway ikev2-gw-ipv4-cert aaa access-profile ikev2-access-profile-1 config-
  payload-password juniper123
8 set security ike gateway ikev2-gw-ipv4-cert version v2-only
9 set security ike gateway ikev2-gw-ipv4-cert fragmentation size 1280
```

Correspondingly, the IPv6 gateway is configured with the address 2a01:5f0:1025:1::173:

```
1 set security ike gateway ikev2-gw-ipv6-cert ike-policy ikev2-policy-cert
2 set security ike gateway ikev2-gw-ipv6-cert dynamic user-at-hostname "@cert.auth"
3 set security ike gateway ikev2-gw-ipv6-cert dynamic ike-user-type group-ike-id
4 set security ike gateway ikev2-gw-ipv6-cert local-identity hostname p5.jnpr.cz
5 set security ike gateway ikev2-gw-ipv6-cert external-interface ge-0/0/2.0
6 set security ike gateway ikev2-gw-ipv6-cert local-address 2a01:5f0:1025:1::173
7 set security ike gateway ikev2-gw-ipv6-cert aaa access-profile ikev2-access-profile-1 config-
  payload-password juniper123
8 set security ike gateway ikev2-gw-ipv6-cert version v2-only
9 set security ike gateway ikev2-gw-ipv6-cert fragmentation size 1280
```

The IPsec (transform) proposal is the same as that for PSK authentication. Please note that, based on observations, Android does not support Perfect Forward Secrecy (PFS):

```

1 set security ipsec proposal ipsec-proposal-cert protocol esp
2 set security ipsec proposal ipsec-proposal-cert encryption-algorithm aes-256-gcm
3 set security ipsec proposal ipsec-proposal-cert lifetime-seconds 3900
4 set security ipsec policy ipsec-policy-cert proposals ipsec-proposal-cert

```

VPN configuration for the IPv4 endpoint: Based on observations, Android does not accept any remote traffic selectors other than **0.0.0.0/0** and **::/0**, respectively. The difference between the PSK and certificate groups is the use of different st0 logical units:

```

1 set security ipsec vpn ipsec-vpn-ipv4-cert bind-interface st0.2
2 set security ipsec vpn ipsec-vpn-ipv4-cert ike gateway ikev2-gw-ipv4-cert
3 set security ipsec vpn ipsec-vpn-ipv4-cert ike ipsec-policy ipsec-policy-cert
4 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
5 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
6 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv6 local-ip ::/0
7 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv6 remote-ip ::/0

```

IPv6 configuration corresponding to the above IPv4 VPN:

```

1 set security ipsec vpn ipsec-vpn-ipv6-cert bind-interface st0.3
2 set security ipsec vpn ipsec-vpn-ipv6-cert df-bit clear
3 set security ipsec vpn ipsec-vpn-ipv6-cert ike gateway ikev2-gw-ipv6-cert
4 set security ipsec vpn ipsec-vpn-ipv6-cert ike ipsec-policy ipsec-policy-cert
5 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
6 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
7 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv6 local-ip ::/0
8 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv6 remote-ip ::/0

```

The assignment of tunnel interfaces for both IPv4 and IPv6 to zones and routing instances is as follows:

```

1 set security zones security-zone vpn interfaces st0.2
2 set security zones security-zone vpn interfaces st0.3
3 set interfaces st0 unit 2 description ipsec-vpn-ipv4-cert
4 set interfaces st0 unit 2 family inet
5 set interfaces st0 unit 2 family inet6
6 set interfaces st0 unit 3 description ipsec-vpn-ipv6-cert
7 set interfaces st0 unit 3 family inet
8 set interfaces st0 unit 3 family inet6
9
10 set routing-instances vr interface st0.2
11 set routing-instances vr interface st0.3

```

Finally, apply both the common and certificate groups:

```

1 top
2
3 set apply-groups dial-up-vpn-common
4 set apply-groups dial-up-vpn-cert
5 commit

```

0.7.4 Android IKEv2 Cert Validation

Normally, a solution like an MDM would handle the distribution of PKI materials. However, for testing purposes, an easy way to retrieve PKI materials is to download them from a secure HTTPS site—both the DER-encoded CA certificate and the PKCS#12 container (*.p12 files) containing the user's private key and certificate (which also includes the CA certificate, but Android does not seem to support the import of the embedded CA certificate). After downloading, import the DER-encoded CA certificate (.cer) and the user's key/certificate from the encrypted PKCS#12 container, via the VPN & app user certificate menu item:

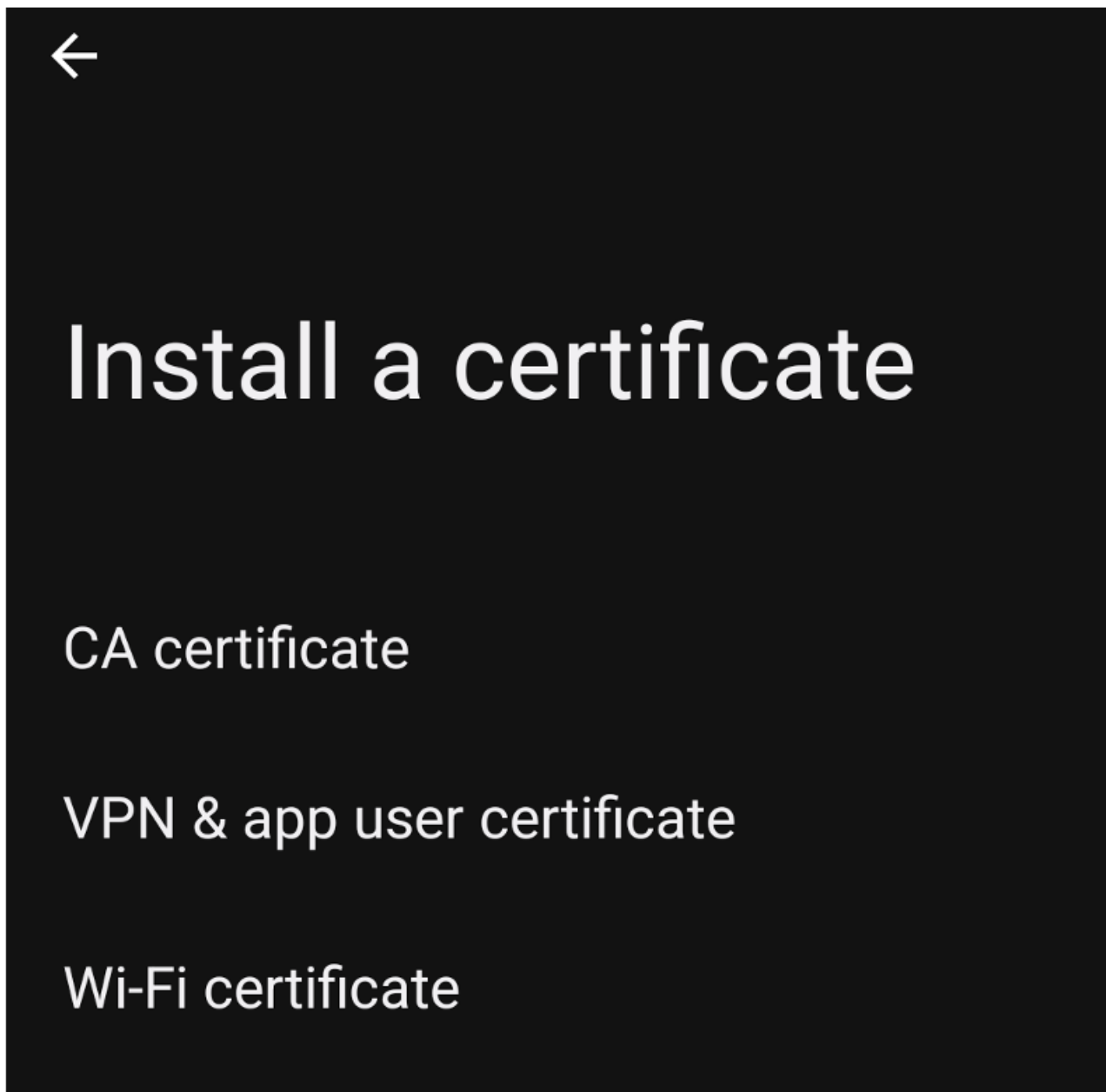


Figure 4: Android Install Certificate Dialogue

In the next step, configure the VPN profile, where a critical part is ensuring that the Server address matches the SRX certificate's DNS SAN and the IPSec identifier corresponds to the client's certificate email SAN.

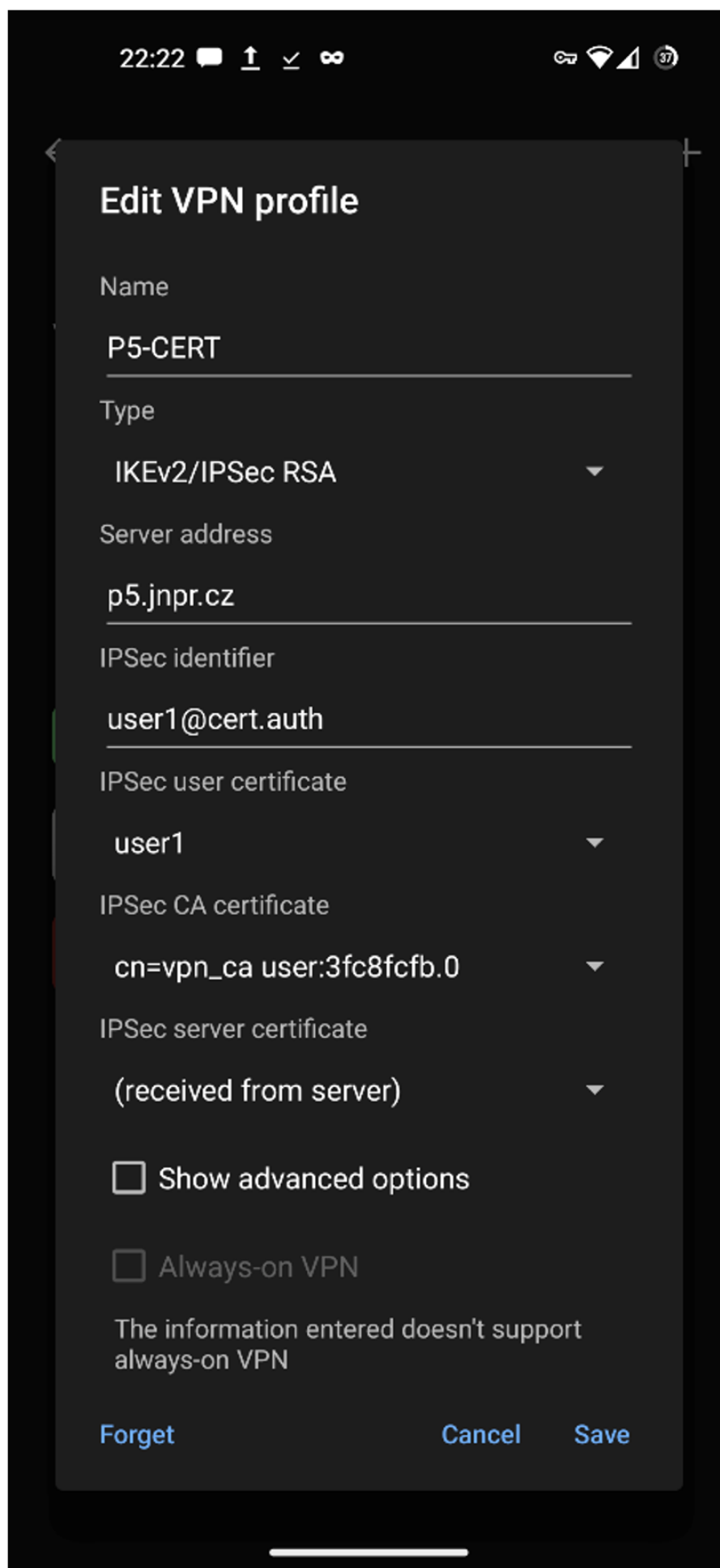


Figure 5: Certificate Authenticated VPN Settings Page 18

Followed by initiating the VPN connection:

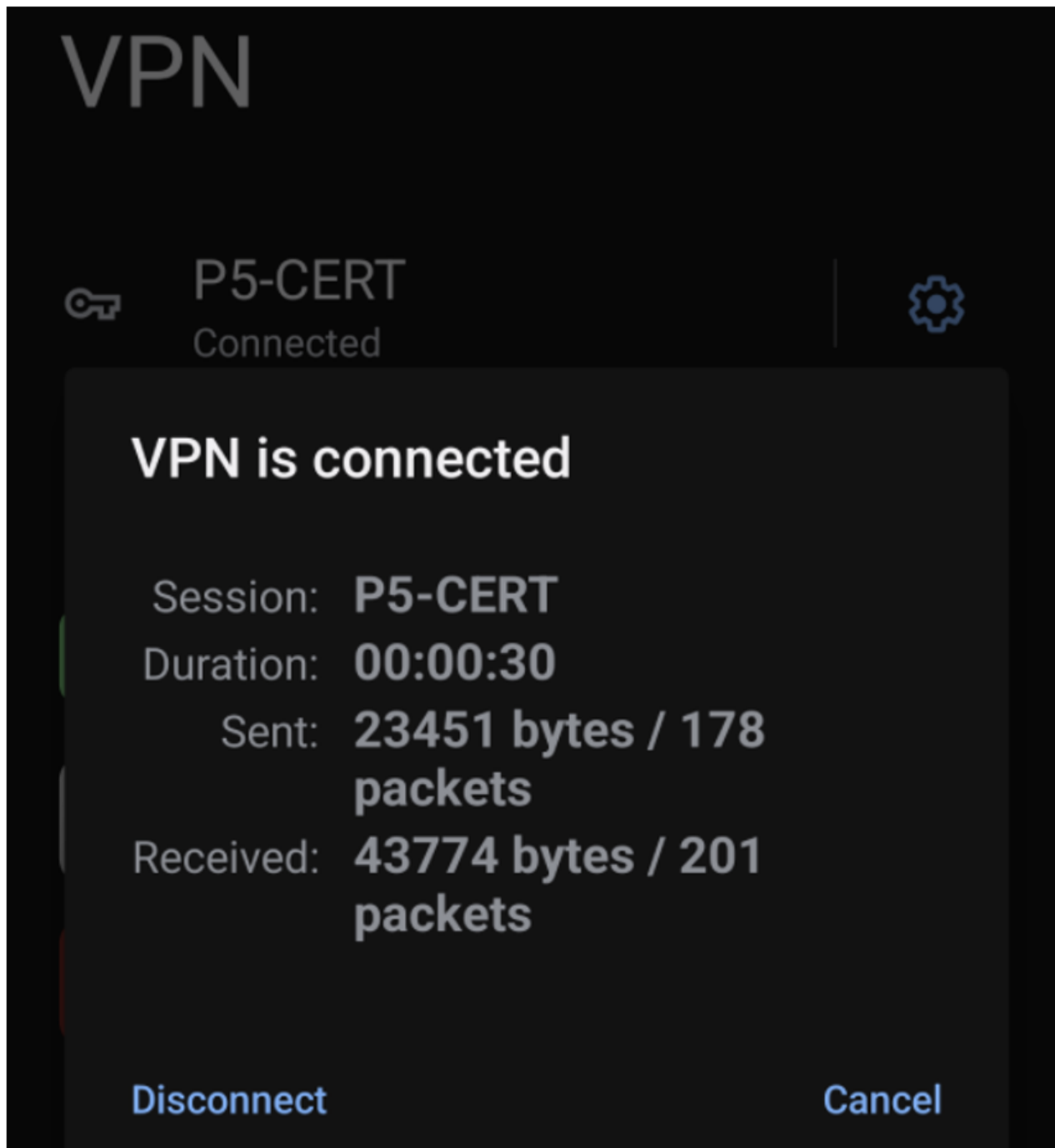


Figure 6: Connected VPN Status

SRX side upon a successful connection:

```
1 > show security ike active-peer detail
2
3 Local gateway interface: ge-0/0/2.0
4 Routing instance: vr
5 Local address: 89.187.136.173, Port: 4500,
6 Local IKE-ID : p5.jnpr.cz
7 Peer address: 90.176.69.153, Port: 17674,
8 Peer IKE-ID : user1@cert.auth
9 AAA username: not available
10 Assigned network attributes:
11 IP Address : 10.1.11.28 , netmask : 255.255.255.0
12 DNS Address : 1.1.1.1 , DNS2 Address : 1.0.0.1
```

```

13 WINS Address : 0.0.0.0 , WINS2 Address : 0.0.0.0
14 Assigned network attributes (IPv6):
15 IP Address : 2a01:5f0:1025:141::1c , prefix : 64
16 DNS Address : 2606:4700:4700::1111 , DNS2 Address : 2606:4700:4700::1001
17
18 Previous Peer address : 0.0.0.0, Port : 0
19 Active IKE SA indexes : 24385
20 IKE SA negotiated : 1
21 IPSec tunnels active : 1, IPSec Tunnel IDs : 500062

```

Corresponding IKED logs are as follows:

```

1 vsrx-p4 iked[17449]: IKE_GATEWAY_ID_VALIDATION_SUCCESS: IKE: Gateway N:ikev2-gw-ipv4-cert L
:89.187.136.173:500 R:90.176.69.153:4412 Successful ike-id:user1@cert.auth U:N/A IKE:IKEv2
Role:R
2
3 vsrx-p4 iked[17449]: IKE_VPN_UP_ALARM_USER: VPN ipsec-vpn-ipv4-cert from 90.176.69.153 is up.
Local-ip: 89.187.136.173, gateway name: ikev2-gw-ipv4-cert, vpn name: ipsec-vpn-ipv4-cert,
tunnel-id: 500063, local tunnel-if: st0.2, remote tunnel-ip: 90.176.69.153, Local IKE-ID: p5
.jnpr.cz, Remote IKE-ID: user1@cert.auth, AAA username: Not-applicable, VR id: root, Traffic
-selector: ts1, Traffic-selector local ID: ipv4(0,0-65535,0.0.0.0-255.255.255.255) ipv6
(0,0-65535,::-ffff:ffff:ffff:ffff:ffff:ffff) , Traffic-selector remote ID: ipv4
(0,0-65535,10.1.11.29-10.1.11.29) ipv6(0,0-65535,2a01:5f0:1025:141::1d-2a01:5f0:1025:141::1d
) , SA Type: dynamic

```

0.7.5 SRX IKEv2 Cert Troubleshooting

Similarly, as with SRX IKEv2 PSK troubleshooting, strongSwan on Linux serves as the tool to validate PKI and VPN SRX head-end settings, offering rich logging and debugging options. The strongSwan installation steps are included in the PSK troubleshooting section and in addition to the PSK authentication scheme, PKI materials need to be transferred to the Linux machine:

PKI materials is placed in the folders as shown below:

Purpose	File path
CA certificate	/etc/swanctl/x509ca/VPN_CA-cert.pem
Private key	/etc/swanctl/private/user2-key.pem
Certificate	/etc/swanctl/x509/user2-cert.pem

Next, the example settings for the IKEv2 VPN in the Linux `/etc/swanctl/swanctl.conf` file for certificate authentication with IPv4 and IPv6 Config Payload addressing are as follows (the remote IPv4 selector is set to specific prefixes to avoid potential loss of remote management connection when using **0.0.0.0/0**). This configuration requires manual VPN tunnel initiation:

```

1 connections {
2   p5 {
3     remote_addrs = p5.jnpr.cz
4     version = 2
5     proposals = aes256gcm16-prfsha384-modp4096
6     keyingtries = 0
7     dpd_delay = 30
8     rekey_time = 8h
9     vips= 0.0.0.0, ::
10    local {
11      auth = pubkey
12      id = user2@cert.auth
13      certs = user2-cert.pem
14    }
15    remote {
16      auth = pubkey
17      id = p5.jnpr.cz

```

```

18     }
19     children {
20         p5 {
21             remote_ts = 10.0.10.0/24, 1.1.1.1/32, ::/0
22             esp_proposals = aes256gcm16
23             start_action = none
24             close_action = none
25             dpd_action = none
26         }
27     }
28 }
29 }

```

Finally, to restart strongSwan, check the PKI materials, and initiate the VPN - the terminal output will indicate a successful VPN connection, followed by a listing of the IPsec Security Associations (SAs):

```

1  # systemctl restart strongswan
2
3  # swanctl --list-certs
4
5  List of X.509 End Entity Certificates
6
7  subject: "CN=user2"
8  issuer: "CN=VPN_CA"
9  validity: not before Feb 20 11:29:39 2025, ok
10             not after Feb 20 11:29:39 2026, ok (expires in 359 days)
11  serial: 04
12  altNames: user2@cert.auth
13  flags: clientAuth
14  authkeyId: ca:f1:a9:0f:70:57:c6:3d:ec:a4:5d:3b:4e:de:54:a4:d6:f8:fc:97
15  subjkeyId: 44:e0:84:c3:9e:85:41:6a:45:87:ee:4b:0c:dd:4c:a2:8f:1f:81:f9
16  pubkey: RSA 4096 bits, has private key
17  keyid: de:6b:14:22:f6:3f:e8:16:a4:5c:b8:6e:97:ea:13:1b:13:8f:af:ee
18  subjkey: 44:e0:84:c3:9e:85:41:6a:45:87:ee:4b:0c:dd:4c:a2:8f:1f:81:f9
19
20  List of X.509 CA Certificates
21
22  subject: "CN=VPN_CA"
23  issuer: "CN=VPN_CA"
24  validity: not before Feb 20 11:24:24 2025, ok
25             not after Feb 18 11:24:24 2035, ok (expires in 3644 days)
26  serial: 01
27  flags: CA CRLSign self-signed
28  pathlen: 0
29  subjkeyId: ca:f1:a9:0f:70:57:c6:3d:ec:a4:5d:3b:4e:de:54:a4:d6:f8:fc:97
30  pubkey: RSA 4096 bits
31  keyid: 57:4e:92:c0:52:b4:c1:20:a9:27:86:a8:5e:e9:05:a4:cb:e7:61:ee
32  subjkey: ca:f1:a9:0f:70:57:c6:3d:ec:a4:5d:3b:4e:de:54:a4:d6:f8:fc:97
33
34  # swanctl --initiate --child p5
35
36  [IKE] initiating IKE_SA p5[1] to 89.187.136.173
37  [ENC] generating IKE_SA_INIT request 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(
38             HASH_ALG) N(REDIR_SUP) ]
39  [NET] sending packet: from 192.168.123.121[500] to 89.187.136.173[500] (712 bytes)
40  [ENC] received packet: from 89.187.136.173[500] to 192.168.123.121[500] (811 bytes)
41  [ENC] parsed IKE_SA_INIT response 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N((40002))
42             CERTREQ V V V V ]
43  [IKE] received NetScreen Technologies vendor ID
44  [ENC] received unknown vendor ID: 09:00:26:89:df:d6:b7:12
45  [IKE] received Juniper SRX vendor ID
46  [ENC] received unknown vendor ID: 67:96:70:50:f1:38:eb:51:de:0c:a6:5f:54:f1:f0:36
47  [CFG] selected proposal: IKE:AES_GCM_16_256/PRF_HMAC_SHA2_384/MODP_4096
48  [IKE] local host is behind NAT, sending keep alives
49  [IKE] received cert request for "CN=VPN_CA"
50  [IKE] sending cert request for "CN=VPN_CA"
51  [IKE] authentication of 'user2@cert.auth' (myself) with RSA signature successful
52  [IKE] sending end entity cert "CN=user2"
53  [IKE] establishing CHILD_SA p5{1}
54  [ENC] generating IKE_AUTH request 1 [ IdI CERT N(INIT_CONTACT) CERTREQ IDr AUTH CPRQ(ADDR ADDR6
55             DNS DNS6) SA TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(EAP_ONLY) N(MSG_ID_SYN_SUP) ]
56  [ENC] splitting IKE message (2206 bytes) into 2 fragments
57  [ENC] generating IKE_AUTH request 1 [ EF(1/2) ]
58  [ENC] generating IKE_AUTH request 1 [ EF(2/2) ]
59  [NET] sending packet: from 192.168.123.121[4500] to 89.187.136.173[4500] (1248 bytes)
60  [NET] sending packet: from 192.168.123.121[4500] to 89.187.136.173[4500] (1023 bytes)
61  [NET] received packet: from 89.187.136.173[4500] to 192.168.123.121[4500] (1227 bytes)
62  [ENC] parsed IKE_AUTH response 1 [ EF(1/2) ]
63  [ENC] received fragment #1 of 2, waiting for complete IKE message

```

```

61 [NET] received packet: from 89.187.136.173[4500] to 192.168.123.121[4500] (1048 bytes)
62 [ENC] parsed IKE_AUTH response 1 [ EF(2/2) ]
63 [ENC] received fragment #2 of 2, reassembled fragmented IKE message (2210 bytes)
64 [ENC] parsed IKE_AUTH response 1 [ IDr CERT AUTH CPRP(ADDR MASK DNS ADDR6 DNS6 DNS6) SA TSi
    TSr ]
65 [IKE] received end entity cert "CN=p5.jnpr.cz"
66 [CFG] using certificate "CN=p5.jnpr.cz"
67 [CFG] using trusted ca certificate "CN=VPN_CA"
68 [CFG] reached self-signed root ca with a path length of 0
69 [CFG] checking certificate status of "CN=p5.jnpr.cz"
70 [CFG] certificate status is not available
71 [IKE] authentication of 'p5.jnpr.cz' with RSA signature successful
72 [CFG] handling INTERNAL_IP4_NETMASK attribute failed
73 [IKE] installing DNS server 1.1.1.1 to /etc/resolv.conf
74 [IKE] installing DNS server 1.0.0.1 to /etc/resolv.conf
75 [IKE] installing DNS server 2606:4700:4700::1111 to /etc/resolv.conf
76 [IKE] installing DNS server 2606:4700:4700::1001 to /etc/resolv.conf
77 [IKE] installing new virtual IP 10.1.11.32
78 [IKE] installing new virtual IP 2a01:5f0:1025:141::20
79 [IKE] IKE_SA p5[1] established between 192.168.123.121[user2@cert.auth]...89.187.136.173[p5.jnpr
    .cz]
80 [IKE] scheduling rekeying in 26412s
81 [IKE] maximum IKE_SA lifetime 29292s
82 [CFG] selected proposal: ESP:AES_GCM_16_256/NO_EXT_SEQ
83 [IKE] CHILD_SA p5{1} established with SPIs c5e3892c_i e23e30ac_o and TS 10.1.11.32/32 2a01:5f0
    :1025:141::20/128 === 10.0.10.0/24 89.187.136.162/32 ::/0
84 initiate completed successfully
85
86 # swanctl --list-sas
87
88 p5: #1, ESTABLISHED, IKEv2, 5d2880919a6e8088_i* ae296a93d8ec9da9_r
89   local 'user2@cert.auth' @ 192.168.123.121[4500] [10.1.11.32 2a01:5f0:1025:141::20]
90   remote 'p5.jnpr.cz' @ 89.187.136.173[4500]
91   AES_GCM_16-256/PRF_HMAC_SHA2_384/MODP_4096
92   established 253s ago, rekeying in 26159s
93   p5: #1, reqid 1, INSTALLED, TUNNEL-in-UDP, ESP:AES_GCM_16-256
94     installed 253s ago, rekeying in 3279s, expires in 3707s
95     in c5e3892c, 0 bytes, 0 packets
96     out e23e30ac, 0 bytes, 0 packets
97     local 10.1.11.32/32 2a01:5f0:1025:141::20/128
98     remote 10.0.10.0/24 89.187.136.162/32 ::/0

```

0.8 Appendix 1 – Complete Configuration

0.8.1 Common Settings

Junos group with common settings for both PSK and Certificate authentication:

```

1  edit groups dial-up-vpn-common
2
3  set apply-flags omit
4  set security flow tcp-mss ipsec-vpn mss 1400
5  set security nat source rule-set untrust from zone vpn
6  set security policies from-zone vpn to-zone trust policy permit-vpn match source-address any
7  set security policies from-zone vpn to-zone trust policy permit-vpn match destination-address
    any
8  set security policies from-zone vpn to-zone trust policy permit-vpn match application junos-icmp
    -ping
9  set security policies from-zone vpn to-zone trust policy permit-vpn then permit
10 set security policies from-zone vpn to-zone untrust policy permit-vpn match source-address any
11 set security policies from-zone vpn to-zone untrust policy permit-vpn match destination-address
    any
12 set security policies from-zone vpn to-zone untrust policy permit-vpn match application any
13 set security policies from-zone vpn to-zone untrust policy permit-vpn then permit
14 set security zones security-zone vpn tcp-rst
15 set security zones security-zone untrust interfaces ge-0/0/2.0 host-inbound-traffic system-
    services ike
16 set access profile ikev2-access-profile-1 authentication-order radius
17 set access profile ikev2-access-profile-1 address-assignment pool ikev2-address-pool-1
18 set access profile ikev2-access-profile-1 address-assignment inet6-pool ikev2-address-pool-2
19 set access profile ikev2-access-profile-1 radius-server 10.0.10.10 secret juniper123
20 set access profile ikev2-access-profile-1 radius-server 10.0.10.10 routing-instance vr
21 set access address-assignment pool ikev2-address-pool-1 family inet network 10.1.11.0/24
22 set access address-assignment pool ikev2-address-pool-1 family inet range ikev2-range-1 low
    10.1.11.2

```

```

23 set access address-assignment pool ikev2-address-pool-1 family inet range ikev2-range-1 high
    10.1.11.200
24 set access address-assignment pool ikev2-address-pool-1 family inet xauth-attributes primary-dns
    1.1.1.1/32
25 set access address-assignment pool ikev2-address-pool-1 family inet xauth-attributes secondary-
    dns 1.0.0.1/32
26 set access address-assignment pool ikev2-address-pool-2 family inet6 prefix 2a01:05f0
    :1025:141::/64
27 set access address-assignment pool ikev2-address-pool-2 family inet6 range ikev2-range-2 low 2
    a01:05f0:1025:141::2/128
28 set access address-assignment pool ikev2-address-pool-2 family inet6 range ikev2-range-2 high 2
    a01:05f0:1025:141::ff/128
29 set access address-assignment pool ikev2-address-pool-2 family inet6 xauth-attributes primary-
    dns-ipv6 2606:4700:4700::1111
30 set access address-assignment pool ikev2-address-pool-2 family inet6 xauth-attributes secondary-
    dns-ipv6 2606:4700:4700::1001

```

0.8.2 PSK authentication

Junos group with configuration for PSK authentication:

```

1 edit groups dial-up-vpn-psk
2
3 set apply-flags omit
4 set security ike proposal ikev2-proposal-psk authentication-method pre-shared-keys
5 set security ike proposal ikev2-proposal-psk dh-group group16
6 set security ike proposal ikev2-proposal-psk encryption-algorithm aes-256-gcm
7 set security ike proposal ikev2-proposal-psk lifetime-seconds 57600
8 set security ike policy ikev2-policy-psk proposals ikev2-proposal-psk
9 set security ike policy ikev2-policy-psk seeded-pre-shared-key ascii-text juniper123
10 set security ike gateway ikev2-gw-ipv4-psk ike-policy ikev2-policy-psk
11 set security ike gateway ikev2-gw-ipv4-psk dynamic user-at-hostname "@psk.auth"
12 set security ike gateway ikev2-gw-ipv4-psk dynamic ike-user-type group-ike-id
13 set security ike gateway ikev2-gw-ipv4-psk local-identity hostname p4.jnpr.cz
14 set security ike gateway ikev2-gw-ipv4-psk external-interface ge-0/0/2.0
15 set security ike gateway ikev2-gw-ipv4-psk local-address 89.187.136.172
16 set security ike gateway ikev2-gw-ipv4-psk aaa access-profile ikev2-access-profile-1 config-
    payload-password juniper123
17 set security ike gateway ikev2-gw-ipv4-psk version v2-only
18 set security ike gateway ikev2-gw-ipv4-psk fragmentation size 1280
19 set security ike gateway ikev2-gw-ipv6-psk ike-policy ikev2-policy-psk
20 set security ike gateway ikev2-gw-ipv6-psk dynamic user-at-hostname "@psk.auth"
21 set security ike gateway ikev2-gw-ipv6-psk dynamic ike-user-type group-ike-id
22 set security ike gateway ikev2-gw-ipv6-psk local-identity hostname p4.jnpr.cz
23 set security ike gateway ikev2-gw-ipv6-psk external-interface ge-0/0/2.0
24 set security ike gateway ikev2-gw-ipv6-psk local-address 2a01:5f0:1025:1::172
25 set security ike gateway ikev2-gw-ipv6-psk aaa access-profile ikev2-access-profile-1 config-
    payload-password juniper123
26 set security ike gateway ikev2-gw-ipv6-psk version v2-only
27 set security ike gateway ikev2-gw-ipv6-psk fragmentation size 1280
28 set security ipsec proposal ipsec-proposal-psk protocol esp
29 set security ipsec proposal ipsec-proposal-psk encryption-algorithm aes-256-gcm
30 set security ipsec proposal ipsec-proposal-psk lifetime-seconds 3900
31 set security ipsec policy ipsec-policy-psk proposals ipsec-proposal-psk
32 set security ipsec vpn ipsec-vpn-ipv4-psk bind-interface st0.0
33 set security ipsec vpn ipsec-vpn-ipv4-psk ike gateway ikev2-gw-ipv4-psk
34 set security ipsec vpn ipsec-vpn-ipv4-psk ike ipsec-policy ipsec-policy-psk
35 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
36 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
37 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv6 local-ip ::/0
38 set security ipsec vpn ipsec-vpn-ipv4-psk traffic-selector ts1 term ipv6 remote-ip ::/0
39 set security ipsec vpn ipsec-vpn-ipv6-psk bind-interface st0.1
40 set security ipsec vpn ipsec-vpn-ipv6-psk ike gateway ikev2-gw-ipv6-psk
41 set security ipsec vpn ipsec-vpn-ipv6-psk ike ipsec-policy ipsec-policy-psk
42 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
43 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
44 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv6 local-ip ::/0
45 set security ipsec vpn ipsec-vpn-ipv6-psk traffic-selector ts1 term ipv6 remote-ip ::/0
46 set security zones security-zone vpn interfaces st0.0
47 set security zones security-zone vpn interfaces st0.1
48 set interfaces st0 unit 0 description ipsec-vpn-ipv4-psk
49 set interfaces st0 unit 0 family inet
50 set interfaces st0 unit 0 family inet6
51 set interfaces st0 unit 1 description ipsec-vpn-ipv6-psk
52 set interfaces st0 unit 1 family inet
53 set interfaces st0 unit 1 family inet6

```

```
54 set routing-instances vr interface st0.0
55 set routing-instances vr interface st0.1
```

0.8.3 Cert authentication

Junos group with configuration for Certificate authentication, please note that PKI configuration is in top hierarchy – when deactivating group, the corresponding CA certificate would be removed:

```
1 edit groups dial-up-vpn-cert
2
3 set apply-flags omit
4 set security ike proposal ikev2-proposal-cert authentication-method rsa-signatures
5 set security ike proposal ikev2-proposal-cert dh-group group16
6 set security ike proposal ikev2-proposal-cert encryption-algorithm aes-256-gcm
7 set security ike proposal ikev2-proposal-cert lifetime-seconds 57600
8 set security ike policy ikev2-policy-cert proposals ikev2-proposal-cert
9 set security ike policy ikev2-policy-cert certificate local-certificate p5-jnpr-cz_VPN_CA
10 set security ike gateway ikev2-gw-ipv4-cert ike-policy ikev2-policy-cert
11 set security ike gateway ikev2-gw-ipv4-cert dynamic user-at-hostname "@cert.auth"
12 set security ike gateway ikev2-gw-ipv4-cert dynamic ike-user-type group-ike-id
13 set security ike gateway ikev2-gw-ipv4-cert local-identity hostname p5.jnpr.cz
14 set security ike gateway ikev2-gw-ipv4-cert external-interface ge-0/0/2.0
15 set security ike gateway ikev2-gw-ipv4-cert local-address 89.187.136.173
16 set security ike gateway ikev2-gw-ipv4-cert aaa access-profile ikev2-access-profile-1 config-
    payload-password juniper123
17 set security ike gateway ikev2-gw-ipv4-cert version v2-only
18 set security ike gateway ikev2-gw-ipv4-cert fragmentation size 1280
19 set security ike gateway ikev2-gw-ipv6-cert ike-policy ikev2-policy-cert
20 set security ike gateway ikev2-gw-ipv6-cert dynamic user-at-hostname "@cert.auth"
21 set security ike gateway ikev2-gw-ipv6-cert dynamic ike-user-type group-ike-id
22 set security ike gateway ikev2-gw-ipv6-cert local-identity hostname p5.jnpr.cz
23 set security ike gateway ikev2-gw-ipv6-cert external-interface ge-0/0/2.0
24 set security ike gateway ikev2-gw-ipv6-cert local-address 2a01:5f0:1025:1::173
25 set security ike gateway ikev2-gw-ipv6-cert aaa access-profile ikev2-access-profile-1 config-
    payload-password juniper123
26 set security ike gateway ikev2-gw-ipv6-cert version v2-only
27 set security ike gateway ikev2-gw-ipv6-cert fragmentation size 1280
28 set security ipsec proposal ipsec-proposal-cert protocol esp
29 set security ipsec proposal ipsec-proposal-cert encryption-algorithm aes-256-gcm
30 set security ipsec proposal ipsec-proposal-cert lifetime-seconds 3900
31 set security ipsec policy ipsec-policy-cert proposals ipsec-proposal-cert
32 set security ipsec vpn ipsec-vpn-ipv4-cert bind-interface st0.2
33 set security ipsec vpn ipsec-vpn-ipv4-cert ike gateway ikev2-gw-ipv4-cert
34 set security ipsec vpn ipsec-vpn-ipv4-cert ike ipsec-policy ipsec-policy-cert
35 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
36 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
37 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv6 local-ip ::/0
38 set security ipsec vpn ipsec-vpn-ipv4-cert traffic-selector ts1 term ipv6 remote-ip ::/0
39 set security ipsec vpn ipsec-vpn-ipv6-cert bind-interface st0.3
40 set security ipsec vpn ipsec-vpn-ipv6-cert df-bit clear
41 set security ipsec vpn ipsec-vpn-ipv6-cert ike gateway ikev2-gw-ipv6-cert
42 set security ipsec vpn ipsec-vpn-ipv6-cert ike ipsec-policy ipsec-policy-cert
43 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv4 local-ip 0.0.0.0/0
44 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv4 remote-ip 0.0.0.0/0
45 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv6 local-ip ::/0
46 set security ipsec vpn ipsec-vpn-ipv6-cert traffic-selector ts1 term ipv6 remote-ip ::/0
47 set security zones security-zone vpn interfaces st0.2
48 set security zones security-zone vpn interfaces st0.3
49 set interfaces st0 unit 2 description ipsec-vpn-ipv4-cert
50 set interfaces st0 unit 2 family inet
51 set interfaces st0 unit 2 family inet6
52 set interfaces st0 unit 3 description ipsec-vpn-ipv6-cert
53 set interfaces st0 unit 3 family inet
54 set interfaces st0 unit 3 family inet6
55 set routing-instances vr interface st0.2
56 set routing-instances vr interface st0.3
```

0.8.4 Top-level Config Sample

Sample configuration for basic infrastructure in top level hierarchy, an important section is PKI required for group with certificate authentication:

```

1 set apply-groups dial-up-vpn-common
2 set apply-groups dial-up-vpn-psk
3 set apply-groups dial-up-vpn-cert
4 set system host-name vsrx-p4
5 set system services netconf ssh
6 set system services ssh root-login allow
7 set system services ssh sftp-server
8 set system services ssh client-alive-interval 120
9 set system time-zone Europe/Prague
10 set system name-server 10.0.0.1
11 set system syslog host 10.0.10.10 any any
12 set system syslog file messages any any
13 set system syslog file messages match "!iked"
14 set system syslog file messages archive size 5m
15 set system syslog file messages archive files 4
16 set system syslog file vpn any any
17 set system syslog file vpn match iked
18 set system syslog file vpn archive size 5m
19 set system syslog file vpn archive files 4
20 set system ntp server 10.0.0.1
21 set services application-identification
22 set security log mode stream
23 set security log report
24 set security log source-address 10.0.10.1
25 set security log stream host format sd-syslog
26 set security log stream host category all
27 set security log stream host host 10.0.10.10
28 set security log stream host host routing-instance vr
29 set security pki ca-profile VPN_CA ca-identity VPN_CA
30 set security pki ca-profile VPN_CA revocation-check disable
31 set security alg h323 disable
32 set security alg mgcp disable
33 set security alg msrpc disable
34 set security alg sunrpc disable
35 set security alg rtsp disable
36 set security alg sccp disable
37 set security alg sip disable
38 set security alg talk disable
39 set security alg tftp disable
40 set security alg pptp disable
41 set security flow tcp-session strict-syn-check
42 set security flow packet-log enable
43 set security flow packet-log throttle-interval 1024
44 set security flow packet-log packet-filter tcp protocol tcp
45 set security nat source rule-set untrust from zone trust
46 set security nat source rule-set untrust to zone untrust
47 set security nat source rule-set untrust rule untrust match source-address 0.0.0.0/0
48 set security nat source rule-set untrust rule untrust match destination-address 0.0.0.0/0
49 set security nat source rule-set untrust rule untrust then source-nat interface
50 set security forwarding-process enhanced-services-mode
51 set security policies from-zone trust to-zone untrust policy permit match source-address any
52 set security policies from-zone trust to-zone untrust policy permit match destination-address any
53 set security policies from-zone trust to-zone untrust policy permit match application any
54 set security policies from-zone trust to-zone untrust policy permit then permit
55 set security policies from-zone trust to-zone untrust policy permit then log session-close
56 set security policies global policy drop-log match source-address any
57 set security policies global policy drop-log match destination-address any
58 set security policies global policy drop-log match application any
59 set security policies global policy drop-log then deny
60 set security policies global policy drop-log then log session-init
61 set security zones security-zone trust tcp-rst
62 set security zones security-zone trust interfaces ge-0/0/0.0 host-inbound-traffic system-services ping
63 set security zones security-zone untrust interfaces ge-0/0/2.0 host-inbound-traffic system-services ping
64 set interfaces ge-0/0/0 unit 0 description trust
65 set interfaces ge-0/0/0 unit 0 family inet address 10.0.10.1/24
66 set interfaces ge-0/0/0 unit 0 family inet6 address 2a01:05f0:1025:140::1/64
67 set interfaces ge-0/0/2 unit 0 description untrust
68 set interfaces ge-0/0/2 unit 0 family inet address 89.187.136.172/27
69 set interfaces ge-0/0/2 unit 0 family inet address 89.187.136.173/27
70 set interfaces ge-0/0/2 unit 0 family inet6 address 2a01:05f0:1025:1::172/64
71 set interfaces ge-0/0/2 unit 0 family inet6 address 2a01:05f0:1025:1::173/64
72 set interfaces fxp0 unit 0 family inet address 10.0.10.2/24
73 set routing-instances vr instance-type virtual-router
74 set routing-instances vr routing-options rib vr.inet6.0 static route ::/0 next-hop 2a01:05f0:1025:1::1
75 set routing-instances vr routing-options static route 0.0.0.0/0 next-hop 89.187.136.161
76 set routing-instances vr interface ge-0/0/0.0

```

```
77 set routing-instances vr interface ge-0/0/2.0
78 set routing-options static route 0.0.0.0/0 next-hop 10.0.10.1
79 set routing-options static route 10.0.0.0/24 next-hop 10.0.10.10
```

0.9 Conclusion

This article documents the basics of VPN connectivity from Android to the Juniper SRX and is essentially meant as a cornerstone for developing performant, scalable, and secure solutions, where high availability is also likely to be taken into account, such as solutions based on Multi-Node High Availability. As a side effect, dial-up VPN from Linux systems is covered as well; although from a troubleshooting perspective, it is still perfectly usable in practice. Two basic schemes were considered: PSK for basic scenarios and PKI for advanced ones. It also turned out that there might be some Android vendor specifics, such as Samsung offering EAP-TLS and different crypto options. The conclusion is that every scenario will need to be thoroughly tested and evaluated based on Android branding and versions.

Sidenote: Outside of JSC, Juniper support will not engage with client-side issues.

0.10 Useful links

- <https://www.juniper.net/documentation/product/us/en/vsrx/>
- <https://www.juniper.net/us/en/dm/download-next-gen-vsrx-firewall-trial.html>
- <https://www.juniper.net/documentation/us/en/software/junos/vpn-ipsec/topics/topic-map/security-vpns-for-ikev2.html#id-example-configuring-the-srx-series-for-pico-cell-provisioning-with-ikev2-configuration-payload>
- <https://source.android.com/docs/core/ota/modular-system/ipsec>
- https://github.com/JNPRAutomate/SRX_VPN_demo_CA
- <https://strongswan.org/>
- <https://juniper.github.io/techposts/srx-autovpn-psk-with-linux-strongswan/article>
- <https://e.foundation/>
- <https://www.freeradius.org/>

0.11 Glossary

- AES: Advanced Encryption Standard
- API: Application Programming Interface
- CA: Certificate Authority
- CBC: Cipher Block Chaining
- CRL: Certificate Revocation List
- DER: Distinguished Encoding Rules
- DH: Diffie Helman
- DNS: Domain Name System
- EAP: Extensible Authentication Protocol
- ECDSA: Elliptic Curve Digital Signature Algorithm
- FQDN: Fully Qualified Domain Name
- GCM: Galois/Counter Mode

- HMAC: Hash-Based Message Authentication Code
- IKE: Internet Key Exchange
- IPSEC: Internet Protocol Security
- JSC: Juniper Secure Connect
- LDAP: Lightweight Directory Access Protocol
- MDM: Mobile Device Management
- MSCHAP: Microsoft Challenge Handshake Authentication Protocol
- MSS: Maximum Segment Size
- MTU: Maximum Transfer Unit
- NAT: Network Address Translation
- OSS: Open-Source Software
- PKCS: Public Key Cryptography Standards
- PKI: Public Key Infrastructure
- PSK: Pre-Shared Key
- RADIUS: Remote Access Dial-In User Service
- SA: Security Association
- SHA: Secure Hash Algorithm
- SQL: Structured Query Language
- TLS: Transport Layer Security
- VPN: Virtual Private Network

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DC – AI – Routing – Switching – Security – Wireless – Automation

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